

# Andrés Aradillas Fernández

Cambridge, MA | (814) 826-5231 | aradilla@mit.edu | aaf101.github.io

## Education

### Massachusetts Institute of Technology (MIT)

*PhD in Economics & Statistics*

Major fields: Econometrics, Industrial Organization | Minor fields: Economic Theory, Computation & Statistics

**Aug 2025 – Present**

*Cambridge, MA*

### Cornell University, College of Arts & Sciences

*Bachelor of Arts with Distinction in All Subjects, Mathematics, Cum Laude*

**Aug 2021 – May 2025**

*Ithaca, NY*

Cumulative GPA: 4.033/4.3 | Honors & Awards: Phi Beta Kappa • Dean's List • ODE Economic Honors Society

## Research & Professional Experience

### Research Assistant, MIT

**May – Aug 2026**

*Professor Victor Chernozhkov*

- Developed causal inference methods for debiased machine learning under AI-learned representations.

### Research Assistant, MIT

**Jan 2026**

*Professor Isaiah Andrews*

- Studied whether bond markets price AI announcements, and constructed a macro-finance model of market responses to AI releases.

### Junior Summer Analyst, Federal Reserve Bank of New York

**Jun 2024 – Aug 2024**

*Financial Stability team, Research & Statistics Group; research under Asani Sarkar*

- Analyzed and compiled firm-level financial data for an empirical study of systemic risk across U.S. nonbank financial institutions.
- Implemented and extended MATLAB code constructing and analyzing sector-level systemic-risk measures; resulting work featured in official New York Fed publication.

### Research Assistant, Econometrics Lab at Cornell University

**Aug 2024 – May 2025**

*Professors José Luis (Pepe) Montiel Olea, Jörg Stoye, and Chen Qiu*

- Developed theoretical and computational methods for optimal decision-making and hypothesis testing for causal inference under noisy signals.
- Adapted online-learning algorithms to solve minimax problems in both econometrics and statistical decision theory.

### Research Assistant, Cornell University

**Oct 2022 – May 2025**

*Professors Francesca Molinari and Levon Barseghyan*

- Developed theoretical results for identification of choice set-dependent decision models and wrote Python simulations of discrete consumer choice with choice set-distorted preferences and choice set-dependent limited selection.

### Visiting Research Intern, Columbia Business School

**May 2023 – Jul 2023**

*Research Assistant to Professor Jing Dong*

- Constructed novel stochastic models of AI-assisted triage based on prior queueing theory work, and performed stress tests and sensitivity analysis in Python.

**Additional Research Experience:** Harvard Business School PRIMO Fellow (2022); Cornell Labor Dynamics Institute (2022–2023). Worked across R, Stata, MATLAB, Python, Git, synthetic control methods, and refined empirical replication methods used in top economics journals.

## Selected Research

*Epsilon-Minimax Solutions of Statistical Decision Problems*

*R&R, Journal of Econometrics*

with José Blanchet, José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Lezhi Tan

- Develop computational methods for approximately solving minimax statistical decision problems using online learning algorithms.

*Pragmatic DML with AI-Learned Representations*

with Victor Chernozhukov, Carlos Cinelli, Sven Klaassen, Whitney Newey, Martin Spindler, Jan Teichert-Kluge, and Suhas Vijaykumar

- Construct a causal inference framework for working with AI-learned representations that can incorporate text and image data into a broad class of estimands.

*Approximate Least Favorable Distributions and Nearly Optimal Tests via Stochastic Mirror Descent*

with José Blanchet, José Luis Montiel Olea, Chen Qiu, Jörg Stoye, and Lezhi Tan

- Develop stochastic optimization algorithms for large statistical decision problems using mirror descent to approximate least-favorable distributions and optimal tests.

## Technical Skills

**Programming:** Python, R, MATLAB, Stata, SQL | **ML/Data:** PyTorch, scikit-learn, NumPy, pandas | **Tools:** Git, Docker, LaTeX